Sunjae Lee

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Research Interests

My research focuses on contributing to the realization of a **magnetic-confinement fusion reactor.** Especially, I'm interested in these specific topics : 1) **kinetic MHD** in 2) **negative triangularity(NT)**, and 3) development of **AI-based high-performance simulations**.

Education

Seoul National University *Mar 2021 – Jul 2026*
College of Liberal Studies / Dept. of Nuclear Engineering Seoul, Korea

- B.S. in **Computer Science & Engineering** and **Physics** *Feb 2026*
- M.S. in **Nuclear Engineering**, enrolled Mar 2026; withdrew in Jul 2026
- **Physics Thesis:** Kinetic MHD stability limit of Low-n mode in Negative triangularity Plasma
- **Computer Science Thesis:** Developing an IMAS-Compatible Database Platform for VEST and Its Application

Columbia University in the City of New York *Fall 2026 – Expected Fall 2031*
Applied Physics and Applied Mathematics (APAM) New York, NY, USA

- M.S./Ph.D. in **Applied Physics (Plasma Physics)**

Research Experiences

Columbia Fusion Research Center, Columbia University *Fall 2026 – Present*
Graduate Research Student New York, NY, USA

Advisor: Prof. Carlos Paz-Soldan

- Decomposed the ideal-MHD δW into shear, parallel-current, and pressure drives for DIII-D NT/PT

Dept. of Nuclear Engineering, Seoul National University *Sep 2024 – Present*
Undergraduate Research Intern, 3DST Lab Seoul, Korea

3 Dimensional Stellarator & Tokamak Lab - Advisor: Prof. Jong-Kyu Park

- Conducted stability analysis of NT using ideal, drift-kinetic, and full kinetic MHD models with GPEC suite
- Found NT has stronger kinetic stabilization effect due to closed-orbit trapped particle's resonance
- Identified a second-stability region of the $n = 1$ in DIII-D NT and analyzed the effect of magnetic shear

Princeton Plasma Physics Laboratory (PPPL) *Jul 2025 – Aug 2025*
Research Student, SNU–PPPL Summer Collaboration Princeton, NJ, USA

- Participated in the PPPL–SNU Summer Research Collaboration
- Learned and applied the BALOO code for edge-stability analysis in the K-DEMO project and NT

Dept. of Nuclear Engineering, Seoul National University *Sep 2023 – Aug 2024*
Undergraduate Research Intern, NUPLEX Lab Seoul, Korea

Seoul National University Plasma & Ion Beam Laboratory - Advisor: Prof. Y. S. Hwang

- Designed and implemented a database for the Versatile Experiment Spherical Torus(VEST) using the IMAS–HSDS data architecture
- Developed a post-processing tool (VAFT) for automated analysis of experimental diagnostic shot data
- Explored operational space for the first time, identifying performance limits from current-driven instabilities

PUBLICATIONS

J: Journal, C: Conference

[J.1] Hong-Sik Yun†, **Sunjae Lee**†, Laurent Jung, et al. (2025). **Developing an IMAS-Compatible Platform for the University-Level Tokamak VEST and Its Application in Operating Characteristics Analysis**. *Plasma Physics and Controlled Fusion*, Vol. 67, DOI:10.1088/1361-6587/ae1b6a

† These authors contributed equally to this work.

ORALS & POSTERS

P: Poster presentations, O: Oral presentations

[P.1] **S. J. Lee**, J. B. Cho, O. Nelson, C. Paz-Soldan, J.-K. Park. (2025). **Kinetic MHD Stability of Low-n Modes in Negative Triangularity Plasmas**. *67th Annual Meeting of the APS Division of Plasma Physics*, Long Beach, CA, USA (Nov 2025)

[P.2] **S. J. Lee**, J. B. Cho, O. Nelson, C. Paz-Soldan, J.-K. Park. (2025).**Kinetic stability of negative-triangularity plasmas**. *29th Workshop on MHD Stability Control*, Princeton, New Jersey, USA (Jul 2025)

[P.3] **S. J. Lee**, J.-K. Park, H.S. Yun, L. Jung, G. W. Nam, G. G. Seo, J. M. Lee (2025). **Standardized Data Infrastructure for Tokamak: Implementation in VEST (Versatile Experiment spherical tokamak)**. *2025 Spring meeting of the Korean Physics Society*, Seoul, Korea (Apr 2025)

[P.4] **S. J. Lee**, H.S. Yun, Laurent Jung, Jung-Hwa Kim, Y.S. Hwang (2024). **Improvements of database system and analysis suite in VEST**. *3rd International Fusion and Plasma conference (IFPC)*, Seoul, Korea (Jun 2024)

TEACHING EXPERIENCE

Teaching Assistant, Introduction to Applied Mathematics — Columbia University — Fall 2026

Teaching Assistant, VEST Winter Camp — SNU — Feb 3–11, 2026
- Tokamak stability course for undergraduate and graduate students

Teaching Assistant, Computer Architecture — SNU — Fall 2025
- Undergraduate-level course

LEADERSHIP & VOLUNTEER EXPERIENCES

Vice President, Sorijigi (Classical Music Listening Club) – SNU — Sep 2021 – Feb 2023
Volunteer, Habitat for Humanity Korea (Snuhabitat) – SNU — Sep 2021 – Feb 2022
Volunteer, Tail – SNU — Aug 2022 – Dec 2023

SKILLS & ADDITIONAL INFORMATIONS

- **Programming Languages:** Python, C/C++, Fortran, Julia, Linux, CUDA, MATLAB, OpenCL, MPI, OpenMP
- **Simulation Tools:** DCON, GPEC, BALOO, KineticEFITtime
- **TOEFL:** 100/120 (Reading 27, Listening 28, Speaking 24, Writing 21)